

Physical and Thermal Characteristics of Dairy Cattle Manure

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Abstract

Greenhouse and regulated gas emissions from animal waste are naturally mediated by moisture content and temperature. As with soils, emissions from manure could be readily estimated given the physical, hydraulic, and thermal properties are described by models and microbes and nutrients are not limiting factors. The objectives of this study were to measure and model physical, hydraulic, and thermal properties of dairy manure to support advanced modeling of gas and water fluxes in addition to solute, colloid and heat transport. A series of soil science measurement techniques were applied to determine a set of fundamental properties of as-excreted dairy cattle manure. Relationships between manure dielectric permittivity and volumetric water content were obtained using time-domain reflectometry (TDR) and capacitance-based dielectric measurements. The measured water retention characteristic for cattle manure was found to be similar to that of organic peat soil. The unsaturated hydraulic conductivity function of dairy manure was inferred from inverse numerical fitting of laboratory manure evaporation results. The saturated hydraulic conductivity was estimated to be about 200 cm day⁻¹. These simulation results suggest that Richards equation can describe the hydrodynamics taking place in dairy manure relevant to natural drying processes. The thermal properties of dairy manure, including thermal conductivity, thermal diffusivity, and bulk volumetric heat capacity were also determined using three penta-needle heat pulse probes (PHPPs). The thermal conductivities of the dairy manure were found between 0.52 and 0.08 W m⁻¹ °C⁻¹ from saturation to dry conditions. Change of the thermal diffusivity during the manure drying process was observed to be only a small range, approximately from 0.0013 (saturation) to 0.0010 cm² s⁻¹ (dry). The bulk volumetric heat capacity of dairy manure at the saturation point was determined as approximately 3.95 MJ m⁻³ °C⁻¹ and linearly decreased to 0.79 MJ m⁻³ °C⁻¹ for the dry manure sample. The accuracy of the measurements was determined from a comparison of theoretical volumetric water content, estimated from the measured thermal properties with that determined by the capacitance-based dielectric measurement. These data represent a novel and unique contribution for advancing prediction and modeling capabilities of gas emissions from cattle manure, while the uncertainties of the results can be due to the complexity of shrinkage, surface crust formation, and shrinkage cracks.

Keywords: Cattle Manure, Physical Properties, Water Retention, Thermal Properties, Drying Process.

1. Introduction

Livestock manure is widely applied to land in agricultural production as the nutrient and organic matter content of manure is beneficial for plant growth, long-term fertility, and soil structure in agronomic systems (Klop et al., 2012; Schröder et al., 2013). However, runoff and infiltration from feedlots and barnyards, land applied manure and from pastures where livestock are grazing can result in transfer of nutrients, pathogens, pharmaceuticals and organic matter to aqueous systems including both ground and surface waters (Christian et al., 2003; Sharpley et al., 1998). Numerical models are required to simulate complex transformation and translocation processes such as with carbon and nitrogen, which involve both liquid and gas phases. There are a number of these models including large-scale land surface models (Del Grosso et al., 2006; Grosso et al., 2008; Oleson et al., 2008; Parton et al., 1998) and point scale models (Simunek et al., 2006; Toride and Chen, 2011), which are continually being improved as more detail is made available, but physical properties of manure have not been defined for use with these models. An accurate simulation model that can describe solute transport from manure sources at a range of scales is mandatory for estimation of quality and quantity of manure leachate.

Accurately describing fluid flow and transport mechanisms in porous media requires extensive knowledge and understanding of physical, chemical, and microbiological processes and properties. Similar to other porous media, the transport and fate of dissolved nutrients in manure depends on the magnitude and direction of the water flux (e.g., infiltration, runoff), which is primarily influenced by the hydraulic gradient (Hillel, 1998; Jury and Horton, 2004). While Richards equation is widely applied in simulations of the fluid transport in unsaturated porous media, it requires the physical and hydraulic properties as the primary input parameters in the model development. As the accumulation of dairy cattle manure in feedlots, storage areas or in pastures generates large amounts of dissolved solutes mobilized by water transport, the physical and hydraulic properties of cattle manure are key requirements in the model development to accurately describe manure leachate transport mechanisms and response from point to field and feedlot scales.

From a physical perspective, manure is a heterogeneous, polyphasic, disperse porous medium generally consisting of solid, liquid, and gaseous phases. The solid fraction primarily consists of fibrous material, which may include hay, grain, and silage, creating a complex manure matrix (Azevedo, 1974; Sobel, 1966; Spellman and Whiting, 2007). The liquid phase is mostly water, commonly containing dissolved solutes and organic matter. The gas fraction occupies the empty pores or void space. The manure matrix determines the geometric characteristics of the empty pores that play an important role in the transport of the water and gases (Hillel, 1998; Horn and Smucker, 2005; Jury and Horton, 2004).

Livestock manure is responsible for approximately 7.5% of methane (CH_4) and 4.7% of nitrous oxide (N_2O) emissions in the US (United States Department of State, 2010). Microbial activity is a key factor for formation of gaseous compounds in manure, the magnitude of gas exchange between manure and the atmosphere largely depends on manure physical characteristics. Microbial metabolism as well as population dynamics (e.g., composition and density) are dramatically influenced by manure temperature (Miller, 1992). However, manure moisture content had a greater influence on microbial activity in the manure composting processes than does temperature (Liang et al., 2003). This is in part due to the competing roles water plays in providing an aqueous environment for microbes while at the same time controlling the rate of

gas exchange (i.e., O₂ supply). The effects of microbial-bacterial growth on the physical and hydraulic characteristics of soil matrix structure in the unsaturated zone are significant (Or et al., 2007). Microbial existence in the soil matrix was found to decrease the evaporative water losses due to an evaporation-retarding barrier, potentially formed by microbes (Chenu and Roberson, 1996; Sutherland, 2001).

Surface crusting has been identified to have a significant impact in decreasing the hydraulic conductivities in infiltration and evaporation processes in soils (Assouline, 2004; Ruan et al., 2001; Touma et al., 2011). Similar principles should apply to surface crust forming processes of cattle manure, which occur at the surface through shrinkage. With drying, capillary forces become more negative, pulling the solid phase fibers closer together, effectively shrinking the pore structure and filling open pore spaces with microbially-generated polymeric substances that have cementation properties. Following crust formation, development of random shrinkage cracks is commonly observed on the cattle manure surface, increasing the difficulty of model predictions. The volume of the sample is known to have an effect on the magnitude of shrinkage cracks (Chertkov, 2013); the larger the manure sample, the more shrinkage cracking is likely to occur. Or (1996) introduced a model of liquid phase sintering of glass compacts to describe wetting induced densification of aggregated soil that could be potentially adopted for crust formation in cattle manure. The model requires information on geometrical and density parameters of cattle manure, which could be obtained from direct measurements while the viscosity of saturated manure, as the unknown parameter, could be estimated using a curve fitting technique. However, more research is needed to evaluate and verify the accuracy of adopting this model to describe crust formation on manure surfaces.

Thermal properties of dairy cattle manure also play a significant role in the drying process. These parameters, including the volumetric heat capacity (C), thermal conductivity (λ), and thermal diffusivity (κ), are used in the heat flow equation to describe the spatial and temporal temperature variations, allowing prediction of the movement of thermal energy and water (Sailor et al., 2008; Scott, 2000). Heat pulse probes have been developed to determine soil thermal properties with high accuracy (Ham and Benson, 2004; Heitman et al., 2003; Knight et al., 2012; Young et al., 2008). The measurement is based on the theory of radial heat dissipation from an infinite line source. Temperature rise associated with a heat pulse applied to the line source is measured at an approximate distance of 6 mm from the line source. The probes normally consist of two parallel needles; one needle providing the heat source and the other containing a thermistor or thermocouple for the temperature measurement.

Heat pulse probes have been successfully used to determine thermal properties and volumetric water content (θ_v), in addition to water flux density of porous media (Gao et al., 2006; Kamai et al., 2008). The penta-needle heat-pulse probes (PHPPs) used in this study employ a novel inverse fitting method for determining soil thermal properties and heat flux density (Sakai et al., 2011; Yang and Jones, 2009; Yang et al., 2013).

A comprehensive literature review clearly revealed that while the biological and chemical decomposition of cattle manure has been widely studied (Gerba and Smith, 2005; Liu et al., 2011; Longhurst et al., 2012; Nennich et al., 2005) with an abundance of reported data, little is known about important physical and thermal properties. As-excreted manures were selected for characterization in this study because they are not contaminated with foreign materials such as

bedding materials or flushed water; therefore considered to be the most reliable data. (Spellman and Whiting, 2007). The objectives of this study were to measure and model physical, hydraulic, and thermal properties of as-excreted dairy manure, that primarily affect flow of liquid water and gas exchange and transport of dissolved constituents. By characterizing physical, hydraulic, and thermal properties of dairy manure using well established analytical models, advanced modeling of greenhouse gas emissions, in addition to water, solute and colloid transport processes can be simulated using analytical and advanced numerical modeling.

2. Theoretical Considerations

Here we define and review critical properties of porous media (manure) and analytical and numerical models used in characterizing and estimating these properties.

2.1 Physical Properties:

Particle density (ρ_s) of organic matter typically ranges between 0.9 to 2.0 g cm⁻³ (Boyd, 1995; Boyd, 2000; Cater et al., 2007; Chen and Avnimelech, 1986). The particle (solid fraction) density of dairy manure was reported within a range of 1.41 to 1.84 g cm⁻³ largely dependent on mineral content of manure (Hafez et al., 1974). The dry bulk density (ρ_b) of organic substrates may range from 0.05 to 0.30 g cm⁻³ (Chen and Avnimelech, 1986). Due to some swelling and significant shrinkage, variations in the bulk density of manure and other organic materials are normally substantial when compared to particle density. Total porosity (f) in most organic materials is commonly greater than 80% by volume (Chen and Avnimelech, 1986).

2.2 Volumetric Water Content Determination of Dairy Manure:

In this study, the water or moisture content of dairy manure (i.e., the quantity of water contained in a manure sample) is described in terms of volumetric water content (θ_v), which is the volume of water contained within a specified bulk volume of manure, which for dielectric sensor-based measurements is inherently given by the sensor output. A large number of techniques have been used to determine θ_v in porous media. Due to a strong correlation between the dielectric permittivity (K_a) and θ_v , sensors are able to estimate θ_v by measuring the apparent K_a (Davis and Chudobiak, 1975). Two different measurement approaches employing electromagnetic (EM) sensing of K_a were used in this study: 1) Travel time analysis using time domain reflectometry (TDR) and 2) capacitance-based measurement using a commercial dielectric sensor (GS3, Decagon Device Inc., Pullman, WA).

A function to describe the relationship between K_a and θ_v can be developed empirically (Malicki et al., 1996; Schaap et al., 1997; Topp et al., 1980), or based on a physical approach employing dielectric theory, which combines constituent dielectric constants and volume fractions of each component (i.e. solid, liquid, and gas). Additional details on TDR measurements are given in Jones et al. (2002) and Robinson et al. (2003). Schaap et al. (1997) derived an empirical relationship between K_a and θ_v for five different organic forest soils using TDR measurements. The empirical calibration equation based on 505 measurements is given as

$$\theta_v = \left(A \sqrt{K_a} - B \right)^C \quad (1)$$

where $A = 0.133 \pm 0.002$, $B = 0.146 \pm 0.002$, and $C = 0.885 \pm 0.018$.

In addition, four potential error sources from TDR calibrations in organic forest soils were described in Schaap et al. (1997). The error sources include decomposition of organic matter during the measurement, residual water after drying, temperature effects, and shrinking of the samples while drying. Among these sources, shrinkage was found to be the most significant factor affecting reliability of the TDR measurements. Nevertheless, because the corrections for θ_v and K_a were approximately comparable, the shrinkage effects on the calibration parameters were insignificant.

2.3 Water, Solute, and Matric Potentials:

The WP4-T Dewpoint Potentiometer, made by Decagon Devices, Inc., determines water potential (ψ_w) of porous media in the laboratory using the chilled mirror dewpoint technique (Gee et al., 1992; Scanlon et al., 2002), based on fundamental thermodynamic relationships and precise measurements of temperature (Campbell et al., 2007). The value of ψ_w measured by the WP4-T is the sum of the osmotic potential (or solute potential, ψ_s) and matric potential, ψ_m , of the porous medium. As the dewpoint method measures the sum of ψ_s and ψ_m , we estimate the contribution of ψ_s using a correlation between ψ_s and electrical conductivity (EC) of the sample solution, corrected for water content (United States Salinity Laboratory Staff, 1954), given as

$$\psi_s(\theta) = -EC_e \left(\frac{\theta_s}{\theta} \right) \cdot 0.036 \quad (2)$$

where $\psi_s(\theta)$ is the water content dependent osmotic potential [MPa], EC_e is the electrical conductivity of the saturation extract [dS m^{-1}], θ_s/θ is the ratio of saturated and actual water contents [$\text{cm}^3 \text{cm}^{-3}$], and 0.036 is a conversion coefficient [$\text{MPa dS}^{-1} \text{m}$].

2.4 Water Retention Curve:

The experimental water retention data were characterized by fitting the van Genuchten model (van Genuchten, 1980) to each data set using

$$\Theta(\psi_m) = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \left[\frac{1}{1 + (\alpha |\psi_m|)^n} \right]^m \quad (3)$$

where Θ is the degree of saturation, ψ_m is the matric potential [-cm], θ is the volumetric water content [$\text{cm}^3 \text{cm}^{-3}$], θ_s and θ_r are the saturated and residual water contents [$\text{cm}^3 \text{cm}^{-3}$], respectively, while α , m , and n ($m = 1 - 1/n$) are the shape parameters related to the pore-size distribution.

2.5 Hydraulic Conductivity:

Numerical analyses of transient water transport problems including infiltration, redistribution and evaporation have been widely used to determine hydraulic parameters of soils with high

accuracy (Hopmans et al., 2002; Ritter et al., 2003; Schwarzel et al., 2006; Si and Kachanoski, 2000; Šimůnek et al., 1998). Kosugi et al. (2001) effectively used inverse simulations to characterize the unsaturated water flow in four types of forest soils, indicating Richards equation can describe the unsaturated water flow in organic forest floors.

In our study, we used the HYDRUS 1-D software package containing the inverse modeling capability to estimate the saturated hydraulic conductivity and unsaturated hydraulic conductivity function. An objective function, including deviations between measured and simulated variables (i.e. water contents) at different times during manure drying, was minimized. Optimization of the objective function was accomplished using the Levenberg-Marquardt nonlinear minimization (Marquardt, 1963). The unsaturated hydraulic conductivity function was estimated from three evaporation experiments during dairy manure drying using the HYDRUS 1-D inverse modeling function. Details of transient flow parameter optimization techniques are described by Hopmans et al. (2002). The van Genuchten hydraulic conductivity model (van Genuchten, 1980) describing unsaturated hydraulic conductivity is given by

$$K(h) = K_s \Theta^l \left[1 - (1 - \Theta^{1/m})^m \right]^2 \quad (4)$$

$$m = 1 - 1/n, \quad n > 1 \quad (5)$$

where K_s is the saturated hydraulic conductivity [cm day^{-1}], Θ is the degree of saturation, n is the same empirical coefficient used in Eq. (3), affecting the shape of the hydraulic functions, and l is the pore-connectivity parameter. Schaap and Leij (2000) recommended using $l = -1$ for most soil textures. Additional information on the numerical solution of the variably saturated flow equation and parameter optimization is given in Šimunek et al. (2009).

2.6 Thermal Properties:

Manure is composed of solid, liquid and gaseous phases and its thermal conductivity (λ) is dependent primarily on the volume fractions of these components, the size and arrangement of the solid particles, and the interfacial contact between the solid and liquid phases (Jury and Horton, 2004). Thermal diffusivity (κ) is a parameter indicating the rate of change of temperature with time as the result of a thermal gradient. An inverse method for optimizing λ and κ of a porous medium from temperature rise measurements was implemented in the FORTRAN program, INV-WATFLX (Yang and Jones, 2009; Yang et al., 2013). Based on the Newton-Gauss-Levenberg-Marquardt method, the INV-WATFLX code simultaneously fits temperature rise emitted from a central heater needle as sensed by four thermistor needles surrounding the heater. The theory of the analytical solution and implementation of the inverse parameter optimization method are given in Yang and Jones (2009) and Yang et al. (2013).

Bulk volumetric heat capacity (C) is defined as the ratio of λ over κ and was determined by fitting of λ and κ values. The bulk volumetric heat capacity of the dairy manure can also be estimated, based on the volume fraction and density of solid organic matter (som) and water (w) composing the dairy manure (the density of air is negligible in comparison with that of solid organic matter and water), given as

$$C_m = \rho_{som}\theta_{som}c_{som} + \rho_w\theta_wc_w \quad (6)$$

where C_m is the bulk volumetric heat capacity of manure [$\text{J m}^{-3} \text{ }^\circ\text{C}^{-1}$], ρ_{som} and ρ_w are the density of solid organic matter and water, respectively [kg m^{-3}], θ_{som} and θ_w are the volume fraction of solid organic matter and water, respectively [$\text{m}^3 \text{ m}^{-3}$], and c_{som} and c_w are the specific heat capacity per unit mass or specific heat of solid organic matter and water, respectively [$\text{J kg}^{-1} \text{ }^\circ\text{C}^{-1}$]. The specific heat capacity of solid organic matter was estimated to be $1,925 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ (Hillel, 1998; Jury and Horton, 2004).

3. Materials and Methods

Techniques commonly applied for soil analysis were applied in this study to examine physical, hydraulic, and thermal properties of as-excreted dairy cattle manure. First, the relationship between manure volumetric water content (θ_v) and apparent dielectric permittivity (K_a) was determined with time domain reflectometry (TDR) and a commercially available capacitance moisture sensor (GS3, Decagon Devices, Inc., Pullman, WA). Secondly, water potential of dairy manure was measured with the chilled-mirror dewpoint technique (WP4-T, Decagon Devices, Inc., Pullman, WA) to investigate the structural and functional relation between manure volumetric water content and water potential under equilibrium conditions. The osmotic potential of the manure was then estimated and subtracted from the water potential to generate the relationship between volumetric water content and matric potential, which is known as the water characteristic or water retention curve. Third, the saturated hydraulic conductivity (K_s) and the unsaturated hydraulic conductivity function of the dairy manure samples was determined by means of an inverse solution simulation techniques as an alternative to direct measurement. We set out to numerically simulate changes in manure moisture content during the drying process with HYDRUS 1-D (Šimůnek et al., 2008), a software package for simulating transient water movement in one-dimensional variably-saturated media with a robust inverse modeling capability. Lastly, the thermal conductivity, thermal diffusivity, and heat capacity of drying dairy manure was measured with PHPPs to investigate and identify relationships between these thermal properties and the volumetric water content during the drying process.

3.1 Study Farm and Manure Sampling:

Dairy manure samples used in this study were collected from the Utah State University Caine Dairy Teaching and Research Center in Wellsville, UT (central coordinates: $41^\circ 39' 22''$ lat; $111^\circ 53' 57''$ long). The milking cows were young, early lactation Holsteins with an average bodyweight of 771 kg and a growth rate of 0.45 kg d^{-1} (1.00 lb d^{-1}). In a feedlot barn, the milking cows were fed a total mixed ration with an approximate crude protein content of 17% and a dry matter intake (DMI) of $27.67 \text{ kg cow}^{-1} \text{ d}^{-1}$ ($61.0 \text{ lb cow}^{-1} \text{ d}^{-1}$). The average milk production for the herd was $40.82 \text{ kg cow}^{-1} \text{ d}^{-1}$. The manure samples were collected as excreted and mixed together in containers. Composite samples were then placed in one-gallon plastic heavy-duty zip lock bags approximately half full, squeezed to remove excess air, sealed and delivered to the lab directly (Peters et al., 2003).

3.2 TDR and GS3 Sensor Calibration in Dairy Manure:

The liquid state of manure can be described in terms of its water retention as (1) the amount of water in manure (expressed in terms of gravimetric wet or dry weight basis or on a volumetric basis) and (2) the potential energy by which water is held within the manure matrix, defined as manure water potential (ψ). Considering manure as a porous medium, the water-retention or -characteristic function describing this relationship is an important characteristic for modeling most other properties and processes, including water and gas transport processes within the manure.

The relationship between θ_v and K_a of manure was characterized at an ambient temperature of about 22 °C. The TDR sensing system included a TDR cable tester (1502B Metallic Cable Tester, Tektronix Inc., Beaverton, OR) connected via a coaxial multiplexer (SDMX50SP, Campbell Scientific Inc., Logan, UT) to three custom three-rod probes with 0.08-m long, 3.20-mm diameter rods and 12.0-mm rod spacing. The waveforms measured with the Tektronix TDR were captured and interpreted for travel time with WinTDR waveform analysis software (Or et al., 2004) on a personal computer. Three GS3 sensors were connected to a data logger (CR1000, Campbell Scientific Inc., Logan, UT) and a personal computer for monitoring K_a .

Three manure samples were prepared for testing with the TDR system and three samples were prepared for testing with the GS3 sensor arrangement. The manure samples were manually packed under gravity into 1,240 cm³ polyvinyl chloride (PVC) rings of 20.32 cm (8 in.) diameter and 3.81 cm (1.5 in.) height. They were carefully prepared with no external compaction applied to maintain the in-situ condition as much as possible. For both the TDR probes and the GS3 sensors, positioning at the center of the rings while packing the manure samples ensured that the fringing fields associated with each measurement was contained within the manure sample volume (Robinson et al., 2003; Vaz et al., 2013). Each sample was left to dry at room temperature over a period of 30 days. For the TDR system, sample masses were continuously recorded with a high resolution balance (GX-6100, A&D Engineering Inc., San Jose, CA) to determine the gravimetric water content on dry weight basis (θ_{md}). For the GS3 sensor setup, the sample masses were continuously monitored using 10 kg pre-calibrated load cells (ESP-10, Transducer Techniques Inc., Temecula, CA) connected to the Campbell Scientific CR1000 data logger, which interfaced with a personal computer. We took care during packing to ensure that there was no air gap between the sensor needles and the manure samples. At the beginning and end of the experiment, the lengths, widths, and volumes of the manure and its subsequent shrunken state were estimated using the sand displacement method (Boelter, 1962). Ultimately, we fitted the equation used by Schaap et al. (1997) for the organic forest soils to our data measured by both TDR system and GS3 sensor to establish general calibration equations for manure.

The final water contents and bulk densities were determined by drying the manure samples at 70 °C for 48 hours to dry while minimizing oxidation of organic material (Peters et al., 2003; Schaap et al., 1997). We determined the oven-dry bulk density of the dairy manure samples to be 0.15±0.012 g cm⁻³. Hafez et al.(1974) reported air-dry bulk density of dairy manure of 0.23 g cm⁻³, somewhat greater than our oven-dried density, which is not surprising given their air-dry weight likely had more residual water resulting in overestimation of the dry bulk density. The particle density of dairy feces was previously reported as 1.44 g cm⁻³ by Sobel (1966). Hafez et al. (1974) reported the particle density of dairy manure and beef cattle manure as 1.43 and 1.44 g

cm⁻³, respectively. Using the particle density of 1.44 g cm⁻³, the solid phase volume of dairy manure samples can be estimated as approximately 0.104.

3.3 Manure Water Retention Curve:

The chilled mirror WP4-T dewpoint Potentiometer with an accuracy of ± 0.1 MPa from 0 to -10 MPa and $\pm 1\%$ from -10 to -300 MPa was used in our study to measure water potentials in laboratory manure specimens. Prior to every measurement, the WP4-T was turned on for 30 minutes as a recommended warm-up period and calibrated with a verification standard, 0.5 mol kg⁻¹ potassium chloride (KCl) salt solution, at 25 °C. Two measurements were read to ensure that both readings were within the range of -2.22 ± 0.10 MPa according to the recommended calibration and verification procedures (Decagon Device Inc., 2007).

Triplicate mixed cattle manure samples were measured to generate the water retention curve for “as-excreted” dairy cattle manure. In each trial, the suite of water potential measurements included six sub-samples prepared from mixed cattle manure samples. The manure sample were equilibrated at a constant room temperature ($\approx 22^\circ\text{C}$) in a sample holder at various potentials for different amounts of time by allowing the water to evaporate from “as-excreted” conditions until the samples were completely air-dry. The sample holder cup was 1.1-cm in height and 4-cm in diameter, being filled half-full with the bottom of the cup entirely covered with the manure samples. The samples were weighed and then placed into the WP4-T and the water potential was measured and recorded. For each water potential reading, three replicates were recorded. The samples were then removed from the WP4-T sample chamber and immediately covered with the sample holder lid preventing evaporation of water. At the end of each trial, the samples were immediately oven dried (70 °C for 24 h) for determination of the water contents.

We paired the matric potentials with the sample volumetric water content measurements to obtain the water retention curve. A nonlinear regression algorithm (Marquardt, 1963) was performed to estimate the model parameters from the measurement data.

3.4 Estimated K_s and $K(\theta)$ of Dairy Manure using HYDRUS 1-D

The van Genuchten-Mualem model (Mualem, 1976; van Genuchten, 1980) was applied in the HYDRUS 1-D inverse solution simulations with the hydraulic properties, including residual water content (θ_r), saturated water content (θ_s), parameters α and n in the water retention function, while the hydraulic conductivity (K_s) was assigned as the unknown parameter to be solved for. Daily evaporation data and changes in volumetric water content, monitored from three evaporation experiments of dairy manure drying in a constant room temperature ($\approx 22^\circ\text{C}$), previously conducted to develop the GS3 sensor calibration equation, were used in establishment of the inverse solution simulations. The monitored daily evaporation rates (Figure 1) were used as time variable boundary conditions. The volumetric water contents, measured from the GS3 moisture sensors, were input into the simulation as the observed parameter to which the objective function in HYDRUS 1-D fits the water retention parameters and unknown K_s and unsaturated hydraulic conductivity function in order to identify the optimal value of K_s and the unsaturated hydraulic conductivity function.

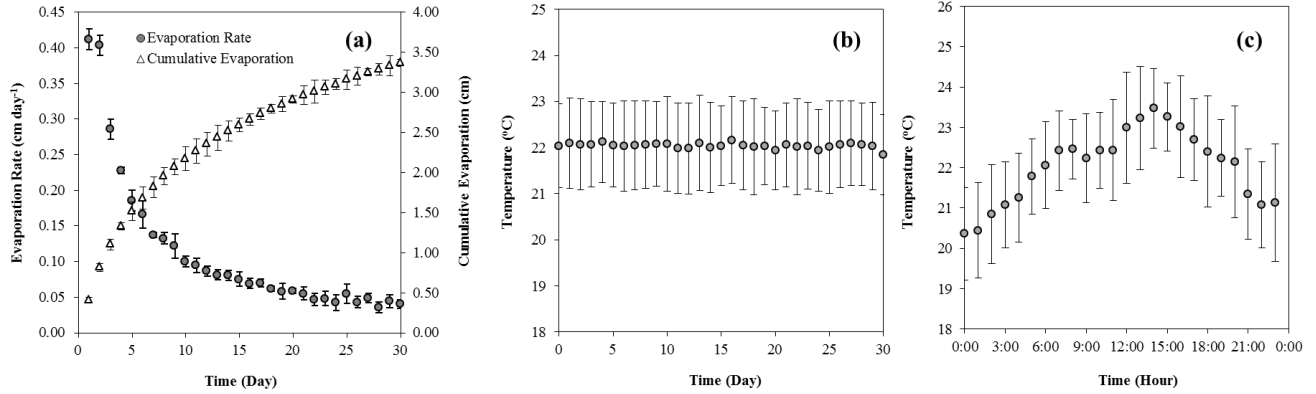


Figure 1. (a) Mean daily evaporation rate (left axis) and cumulative evaporation (right axis) monitored during the evaporation experiments of three dairy manure samples at room temperatures. (b) Daily average of hourly room temperatures. (c) 30-day average of room temperatures recorded on the hour. All error bars denote plus and minus one standard deviation from triplicate sample measurements.

2 3.5 Thermal Properties of Dairy Manure:

3 In this study these three major thermal properties were investigated during the drying process of
 4 manure using PHPPs. The PHPP includes a central heater needle surrounded by an orthogonal
 5 arrangement of four thermistor needles (Figure 2). Based on the Gauss-Newton-Levenberg-
 6 Marquardt method, the thermal properties were determined using an analytical solution that
 7 simultaneously fits time series of temperature measurements from each of the four thermistor
 8 needles. Additional discussions on the PHPP and the inverse fitting method for determination of
 9 the thermal properties are given in Sakai et al.(2011), Yang and Jones (2009), and Yang et al.
 10 (2013).

11 The PHPPs simultaneously determined thermal properties of dairy cattle manure, namely,
 12 thermal conductivity (λ), thermal diffusivity (κ), and volumetric heat capacity (C) during the
 13 laboratory drying process. The PHPP consists of one 2.1-mm (OD) line-source heater needle and
 14 four 1.27-mm (OD) parallel thermistor needles with a physical center to center spacing of 6.5
 15 mm (Figure 2). The two pairs of thermistor needles are orthogonally arranged equidistant from
 16 the heater needle. The precision and stability of the temperature measurements of the PHPPs are
 17 essential for determination of the thermal properties of the manure samples. To evaluate the
 18 precision of the thermistors used in the PHPPs, we used the PHPPs for measuring temperatures
 19 in an insulated container filled with saturated sand with 5-second interval reading for 10 minutes.
 20 The resulting temperatures were consistent, with less than 0.001°C fluctuation.

21 Dairy manure samples were prepared in triplicate for monitoring the thermal properties using the
 22 PHPPs. Each sample was prepared using similar procedures described in the TDR and GS3

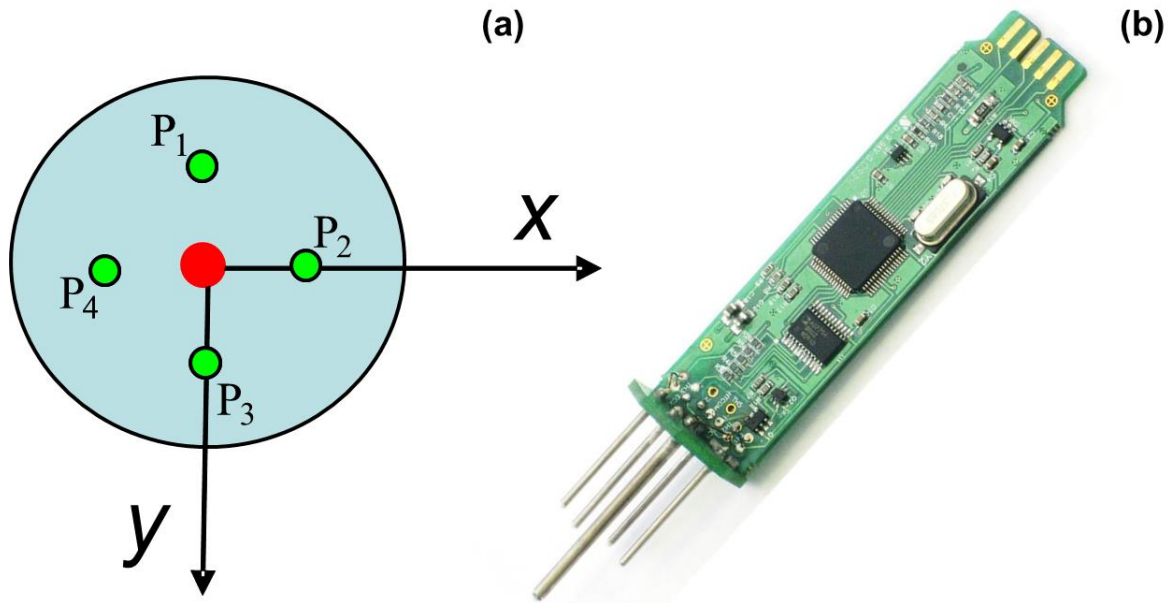


Figure 2. (a) End view of the penta-needle heat pulse probe (PHPP) depicting the location of the heater needle and four thermistors (P_1 , P_2 , P_3 , and P_4). (b) The actual PHPP used for determination of the thermal properties of the dairy manure samples. Prior to the measurements, the electrical components were coated with water resistant epoxy resin.

sensor calibration descriptions. The PHPP was inserted at the middle height of the PVC ring, prior to filling the ring with the manure sample. A Decagon Devices GS3 water content sensor was buried in the sample at the same depth. After the PVC ring was fully filled with the manure sample, the sample then was left to dry in a greenhouse with temperature controls. Attention was given to packing the manure around the sensor to avoid air gaps between the PHPP needles and between the needles of the GS3 sensors. The PHPP and GS3 sensor were connected to and communicated with a data logger (Model CR1000, Campbell Scientific, Inc., Logan, UT) via SDI-12 communications. The PHPP includes an onboard microcontroller, which is programmed to control the heat source needle to generate a heat pulse of 8 seconds and the thermistor needles to measure temperature for a period of 60 seconds. Initial temperature was measured immediately prior to applying the heat input. The microcontroller processes onboard optimization of λ and κ values using the measurements of the temperature rise with additional computations performed on the data logger (Sakai et al., 2011; Yang and Jones, 2009; Yang et al., 2013).

3.6 Statistical Analyses:

The measured data presented in this study were statistically analyzed with the general statistical analysis module of the R statistical software package version 2.14.1 (R Development Core Team, 2011). For all analyses, a p-value of 0.05 or smaller was considered significant.

4. Results and Discussions

4.1 GS3 Sensor Volume of Influence:

Determination of zone or volume of influence for EM sensors (i.e. the volume generating most of the medium response) is essential to better understand the response of a medium to the sensor output. The approximate volume of influence for the GS3 sensor was determined according to the method described in Druyts et al. (2010) and Jones et al. (2005). In brief, the volume of influence for the GS3 sensor was estimated by observing K_a output from the sensor with a variety of probe orientations measured through a range of immersion depths in water i.e. in air, partially submerged in water, and completely submerged in water. Due to the strong contrast between the permittivity of water ($K_a \approx 80$) and air ($K_a \approx 1$), the layer of influence was estimated by initially placing the sensor in the air at a height above the water and gradually submerging the sensor into the water until K_a of 80 was reached at a specific depth of water. Ten independent replicate measurements were recorded for each depth and sensor position arrangement to determine the assay statistics.

The volume or zone of influence of the GS3 sensor was estimated in both axial (vertically along the sensor, y-axis in Figure 3) and radial (perpendicular to the sensor, x-axis and z-axis in Figure 3) components. In normal applications of soil moisture measurements, the axial sensibility establishes the sensor's depth resolution, and the radial sensitivity determines the susceptibility to lateral heterogeneities (Dean et al., 1987). The range and shape of the primary volume of influence predominantly depend on the sensor's physical geometry (Starr and Paltineanu, 2002). We found the axial zone of influence of the GS3 sensor to be 6.50 cm, originated at the inside face of the sensor, and the radial zone of influence to be primarily within 6.30 cm, centered at the middle prong (Figure 3). The volume of influence was then approximated as 400 cm³, 33% greater than specified by the manufacturer. Apparent ranges of sensitivity of the dielectric

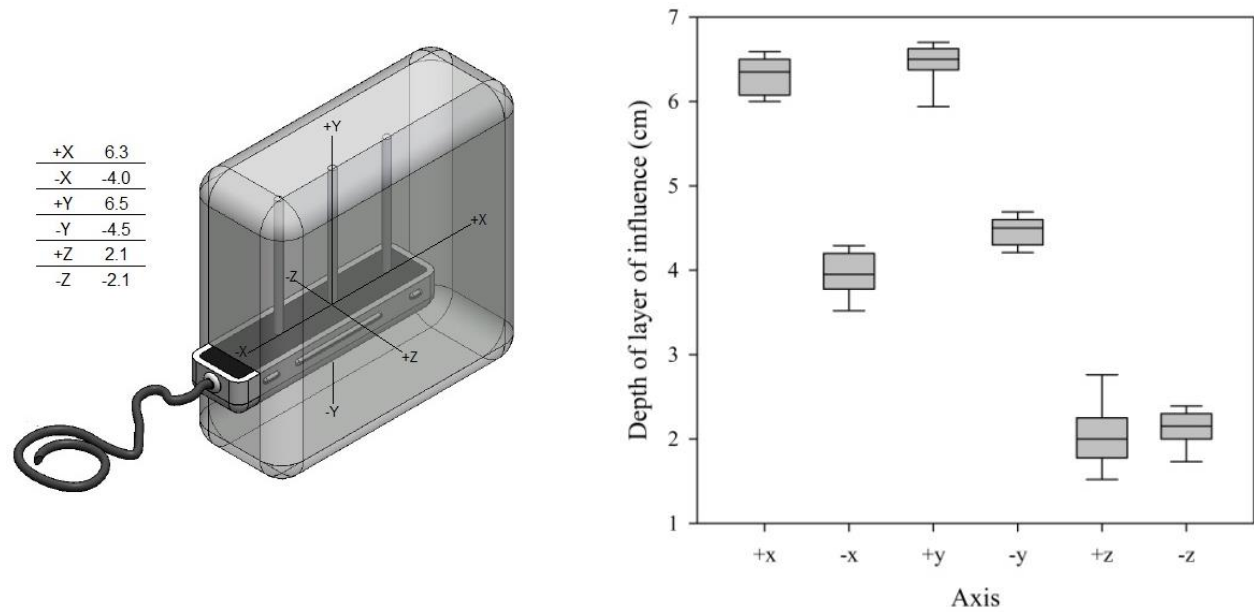


Figure 3. The GS3 sensor and its estimated zone of influence. The dimensions shown on the left are the mean values (in centimeter) from ten replicated independent measurements. The variations of the measurements from ten replicate measurements are shown on the right.

permittivity (K_a) and electrical conductivity (EC) determined with the sensor were similar, due to the linear relationship between these two parameters, programmed in the sensor microprocessor (Decagon Device Inc., 2012). The results suggest the significance of the proper sensor installation to ensure intimate contact between the dairy manure sample and sensor, and the importance in obtaining gravimetric samples within the sensor's zone of primary influence for the sensor calibrations. We note that it is critical to calibrate the sensor in the same installation mode it is anticipated being used in. For example if the sensor will be inserted into the sample surface with the head left above the sample surface, the calibration must be performed similarly. On the other hand, if the sensors are to be buried completely in the sample, as ours were, the calibration should be performed similarly. The reason for this lies in the sensitivity of the GS3 to dielectric of the surrounding medium that extends above/behind the sensor head as seen in Figure 3.

4.2 Cattle Manure Dielectric - Moisture Content Relationships:

The GS3 sensors and TDRs were calibrated in dairy manure to establish a generic calibration equation identifying the correlation between K_a measured with the sensors and θ_v . All manure samples showed considerable shrinkage both in depth and diameter; however, no air gaps between the sensor needles and manure samples were found, confirming the validity of the measurements. Figure 4 presents all measurements performed by both GS3 sensors and TDRs, each with three replicate samples. Overall, the K_a outputs among three dairy manure samples for each measurement technique were highly consistent. The deviation of the measurement between the GS3 sensor and TDR is anticipated owing to the different measurement frequencies (i.e., order of magnitude difference, (Kelleners et al., 2005)) as illustrated in Figure 4. This deviation grows with increasing volumetric water content, which likely results from the high electrical conductivity of the dairy manure (≈ 4.50 dS/m) which increases with the water content. The TDR signal can become completely attenuated as the highly saline manure samples approach saturation (Jones et al., 2002; Mojid et al., 2003).

Table 1 lists the parameters fitted to our measurements and their estimation accuracy, based on the mathematical expression used by Schaap et al. (1997) for organic forest soils. In general, our fitting parameters indicate reliable water content estimates with high R^2 values from both TDR and GS3 sensor measurements. It is worth noting that our calibration methods were performed in a constant room temperature ($\approx 22^\circ\text{C}$), whereas under field conditions diurnal fluctuations in temperature and associated changes in electrical conductivity in addition to variable near-surface water content are common as observed with many methods of measuring soil water (Jones et al., 2005; Or and Wraith, 1999; Starr and Paltineanu, 1998; Wraith and Or, 1999). Techniques for correcting electrical conductivity and temperature sensitivity on capacitance measurements have been reported in several studies (Cobos and Campbell, 2007; Fares et al., 2009; Kelleners et al., 2004; Saito et al., 2013).

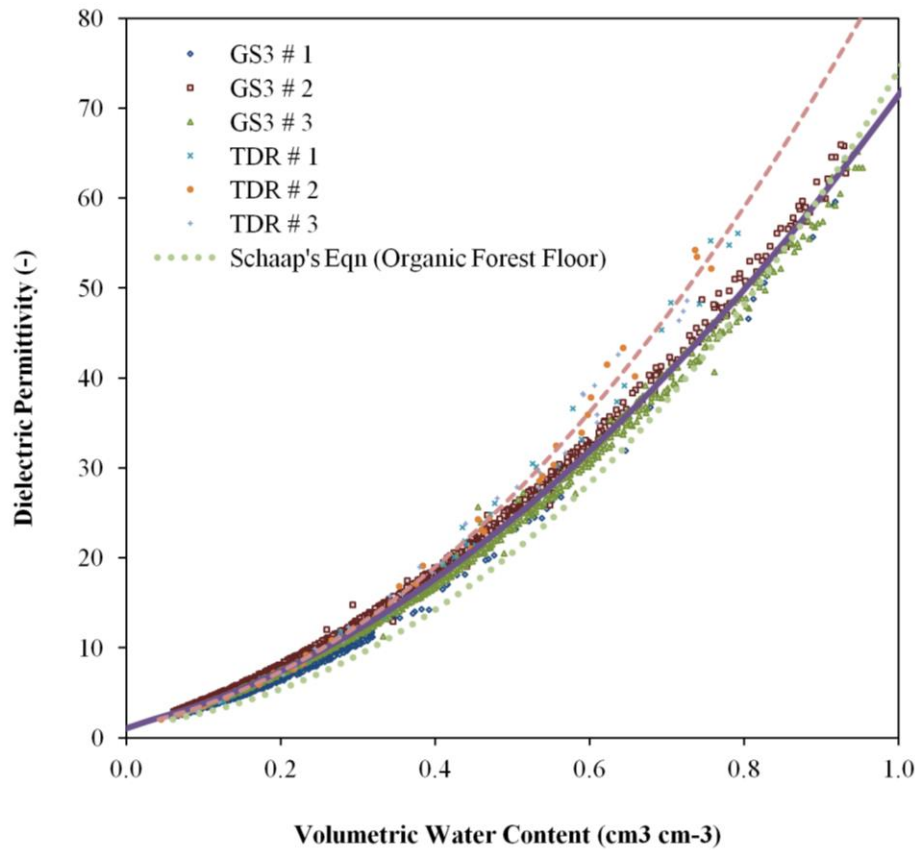


Figure 4. Measured water contents versus dielectric permittivity determined with GS3 sensors (fitted to solid line) and TDR probes (fitted to dashed line). The Schaap et al. (1996) curve for TDR-measured organic forest soils is also shown for reference.

4.3 Dairy Manure Water Retention:

Figure 5(a) illustrates the water retention curve of dairy cattle manure measured with the WP4-T dew point potentiometer. The osmotic potential of the saturation extracts of 18 replicates samples estimated using Eq. (2), were -13.80 m with a standard deviation of -0.54 m, indicating low variability among sample replicates. The osmotic potential contributes substantially to the total water potential of the dairy manure samples and we therefore found it necessary to account for its contribution to the measured water potential and remove its effect yielding only the matric potential used in developing the water retention curves. Due to the high porosity of dairy manure, the saturated water content (θ_s) of the manure samples was much greater than that of mineral soils, which is in agreement with that of other organic materials. Figure 5(b) shows the water retention curve of dairy manure in comparison with *Sphagnum* peat (high bog peat) and reed peat (fen peat) materials (Paivanen, 1973). The relationship between θ_v and ψ_m in organic matter depends on degree of decomposition and botanical composition of residues (Jan et al., 2002).

Table 1. Parametric expression and accuracy of the parameters fitted to the measured data to determine the relationship between K_a and θ_v

Measurement Technique	Coefficients $\theta_v = (A\sqrt{K_a} - B)^C$			n	RMSE (cm ³ cm ⁻³)	R^2
	A	B	C			
GS3 - Dairy manure	0.136	0.150	1.061	2579	0.0125	0.982
TDR - Dairy manure	0.121	0.130	0.990	135	0.0120	0.987
GS3 (Decagon Device Inc., 2012) - Non-mineral soils†	0.118	0.117	1.000	-	-	-
TDR (Schaap et al., 1997) - Organic forest soils	0.133	0.146	0.885	505	-	0.963

† potting soils, perlite, and peat moss at salinities ranging from 0 to greater than 4 dS/m.

The accuracy of the water retention curve in the low matric potential range is limited due to the relatively high osmotic potential of the manure samples near saturation together with diminishing measurement resolution of the WP4-T dewpoint potentiometer near saturation (i.e., 0.01MPa). Measurements near saturation that may be made by using other techniques can determine matric potential in this range (e.g., hanging water column, pressure plate, Tempe Cell, tensiometer). Estimated van Genuchten model parameters derived from the optimization algorithm compared with the previous studies that investigated those for peat soils are listed in Table 2. Schwarzel et al. (2006) applied inverse parameter estimations to determine the hydraulic properties of peat soils of very humified organic peat soils on the surface layer (<15 cm). Da Silva et al. (1993) reported the hydraulic properties of organic peat soils in drying process. These values are nearly the same range of those for the dairy manure samples found in our study. However, Naasz et al. (2005) reported the hydraulic parameters θ_s , α , and n for drying organic peat soils in a different range. The parameters α and n , which are unusually different from other studies, likely resulted from different pedogenetic processes of the soil samples and parameter fitting criteria (Schwarzel et al., 2006).

Because the shape of the water retention curve is dependent on the medium structure, especially in the low matric potential range, it is interesting that the effect of the manure's surface crust formation and shrinkage can potentially change the water retention characteristic by reduction of the total porosity, particularly the volume of the large pores. Consequently, the manure's saturated water content (θ_s) and the initial decrease rate of water content are diminished. Some of the original large pores are forced into the intermediate-size pores due to the shrinkage, creating more intermediate-size pores than the initial condition. While the intermediate-size pores are anticipated to increase, the micro pores remain unaffected, making the water retention curve for manure's surface crust in the high matric potential range unchanged from the original shape. Further investigation is needed to better understand and characterize manure's surface crust and its dynamic properties.

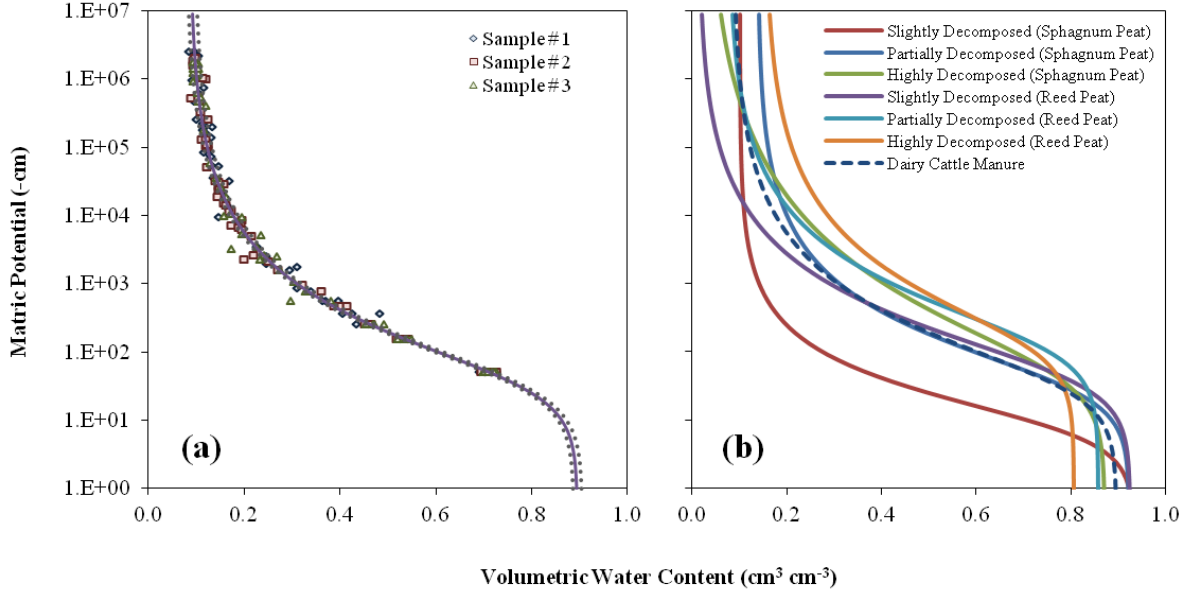


Figure 5. (a) Measured water retention curve of as-excreted cow manure determined by the WP4-T dewpoint potentiometer. The solid line illustrates the van Genuchten hydraulic model with the fitted parameters. The dashed lines show 95% confidence interval of the curve fitting. (b) Water retention curve of dairy manure (dashed line) in comparison with *Sphagnum* and reed peats in different degrees of decomposition (Paivanen, 1973).

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Table 2. Model parameters obtained from fitting the van Genuchten parametric expression to the measurement data compared with the previous studies that investigated those for peat soils

Authors	Medium	van Genuchten model parameters					
		θ_r (cm ³ cm ⁻³)	θ_s (cm ³ cm ⁻³)	α (cm ⁻¹)	n	$m^\dagger$	R^2
This study‡	Cattle Manure	0.0869	0.895	0.027	1.391	0.281	0.997
Schwarzel et al. (2006)	Peat	0	0.880	0.026	1.190	0.160	0.98
Naasz et al.(2005)	Peat	0.3390	0.875	0.0068	10.30	0.180	0.99
Da Silva et al. (1993)	Peat	0	0.901	0.0264	1.390	0.281	0.99

$^\dagger m = 1 - 1/n$, except the study by Naasz et al.(2005) that m was fitted to the measurement data as a model parameter

‡ Number of replicates = 3 with the total number of measurements (N) = 210

4.4 Saturated Hydraulic Conductivity and Unsaturated Hydraulic Conductivity Function:

The inverse simulation with HYDRUS-1D provided estimates for saturated hydraulic conductivity (K_s) and the unsaturated hydraulic conductivity function of dairy manure. The inverse solution yielded good agreement in volumetric water content when compared between the simulated and measured values as illustrated in Figure 6(a) and (b). Table 3 summarizes the estimation of the saturated hydraulic conductivity of dairy manure samples. Figure 6 (c) illustrates the relationship between the volumetric water content (θ_v) and unsaturated hydraulic conductivity for dairy manure, based on the van Genuchten-Mualem model (van Genuchten, 1980).

Table 3. Statistical summary of non-linear regression analysis from estimation of the saturated hydraulic conductivity (K_s) with the HYDRUS-1D inverse solution simulations

Parameter	Value (cm day ⁻¹)
Saturated hydraulic conductivity (K_s)	190
Number of observed volumetric water contents (N)	90
Standard error coefficient (SE)	56.88
Lower 95% confidence limit	81.173
Upper 95% confidence limit	307.20

The hydraulic conductivity function, derived by optimizing changes of the volumetric water contents from three dairy manure samples in the evaporation experiment, effectively characterized the vertical transient water flow in dairy manure in our study. The results support the validity of applying Richards equation to characterize the unsaturated water flow in dairy manure, similar to organic forest (Kosugi et al., 2001) and organic peat soils (Da Silva et al., 1993; Naasz et al., 2005; Schwarzel et al., 2006). On the other hand, Ingram et al. (1974) and Rycroft et al. (1975a; 1975b) observed the “non-Darcian” behavior of organic peat soils, where Darcy’s law is not valid in organic peat soils with a high degree of decomposition in addition to issues of hydrophobicity which complicate things further. Saturated hydraulic conductivities for organic peat soils were reported over a wide range. Schwarzel et al. (2006) estimated K_s of organic peat soils in the surface layer to be 33.50 cm day⁻¹. Naasz et al. (2005) reported the value of K_s of drying organic peat as 3,326.4 cm day⁻¹. Nagare et al. (2013) measured K_s of organic peat in a laboratory using split-container and wax method and found K_s in a range between 2,100 and 31,400 cm day⁻¹.

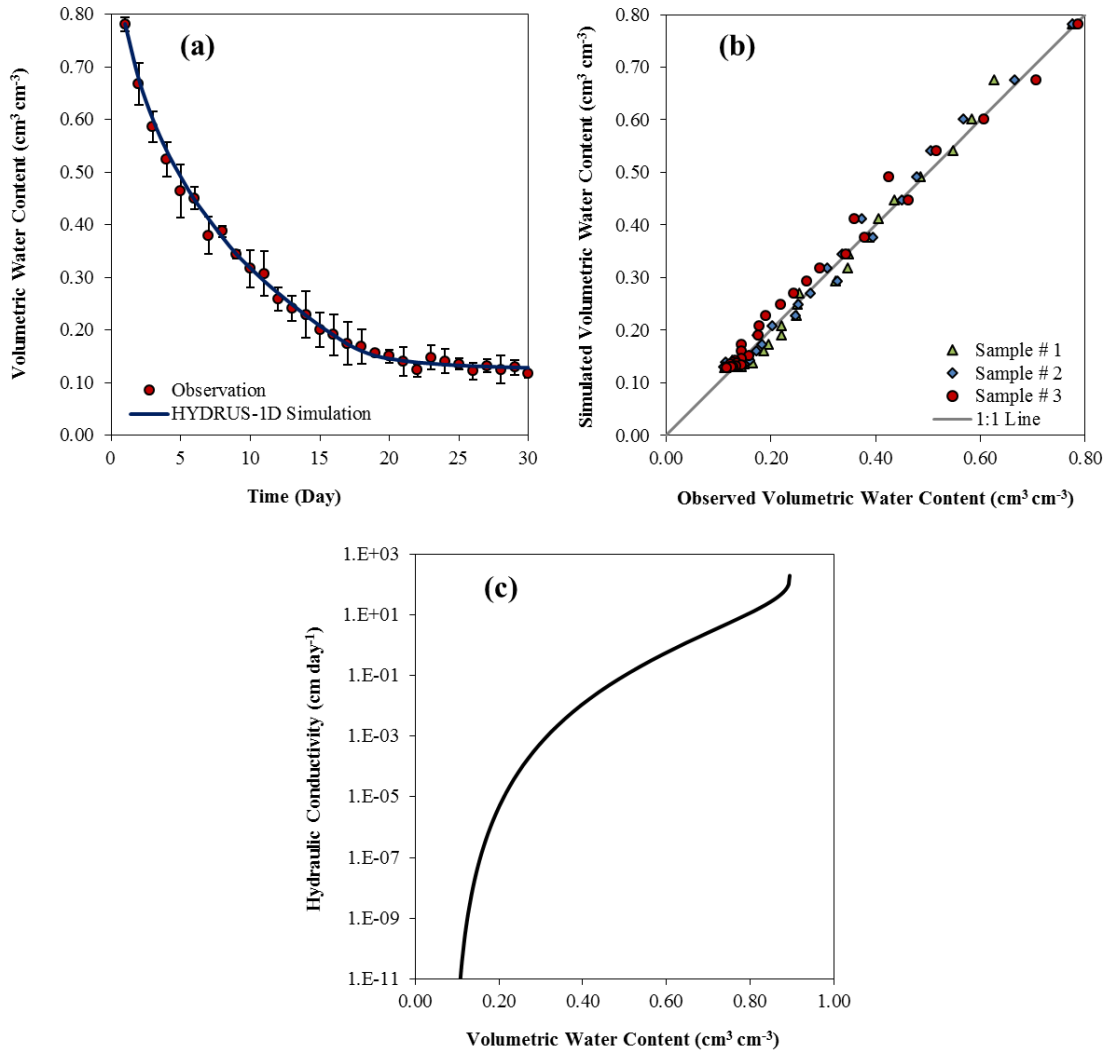


Figure 6. (a) Simulated volumetric water contents fitted to the measurements during the evaporation experiments. The error bars denote plus and minus one standard deviation. (b) Comparison between the measured and simulated volumetric water contents. The $RMSE$ and R^2 value of the model simulation relative to the measured volumetric water content is $0.020 \text{ cm}^3 \text{cm}^{-3}$ and 0.9884 , respectively (c) Unsaturated hydraulic conductivity for dairy manure as a function of water content (θ_v), based on the van Genuchten-Mualem model.

It should be noted that the estimated K_s represents the “effective” saturated hydraulic conductivity of the dairy manure samples. This is complicated by the additional substances in the manure that increase liquid viscosity and form strong bonds upon drying. As-excreted dairy manure is prone to variable surface crust formation and therefore likely has a variation in K_s with drying. Similar to soil crusts, the crust layer of dairy manure contains higher bulk density and lower porosity than the underlying manure due to a higher shrinkage rate together with sodium and total salt contents of manure, which consequently may result in a surface saturated hydraulic conductivity several orders of magnitude less than in the underlying manure (Miller and

Radcliffe, 1992). Another uncertainty that was not considered in our inverse solution simulation for the evaporation experiment is potential changes in pore space geometry due to clogging of pores by gas bubbles and other by-products of organic matter decomposition through anaerobic microbiological processes in manure samples. Prediction of manure surface crust characteristic with associated changes in pore spaces from gas bubbles is difficult because of the random factors in formation and development processes. Further investigations are warranted to more accurately characterize manure surface crust and pore space dynamics in drying and wetting dairy manure.

4.5 Thermal Properties:

Figure 7(a) – (c) illustrates fitted values of λ and κ in addition to computed values of C as a function of θ_v . Table 4 shows the parametric expressions, fitted parameters, and coefficients of determination (R^2) for the relationships between the thermal properties and θ_v . Generally, λ and C decreased linearly with decreasing θ_v (Figure 7(a) and (b)), while κ decreased slowly at the beginning of the drying process, then decreased rapidly once θ_v was below $0.30 \text{ m}^3 \text{ m}^{-3}$ (Figure 7(c)).

The thermal conductivities of the dairy manure sample were found to be between 0.52 and $0.08 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$ from saturation to dry conditions, consistent with the values reported in previous studies (Table 5). The correlation between λ and θ_v , illustrated in Figure 7(a), supports the strong linear relationships of the two parameters, reported in previous studies (Ahn et al., 2009; Chandrakanthi et al., 2005; Nayyeri et al., 2009; Opoku et al., 2006). In addition to θ_v , Nayyeri et al. (2009) demonstrated the first order linear model of the temperature effect on thermal properties of dairy cattle manure.

Table 4. Parametric expressions and parameters describing the thermal conductivity (λ), thermal diffusivity (κ), and volumetric heat capacity (C) as a function of the volumetric water content (θ_v).

Thermal property	Unit	Expression	Parameter			R^2
			a	b	c	
λ	$\text{W m}^{-1} \text{ }^\circ\text{C}^{-1}$	$\lambda = a \cdot \theta_v + b$	0.5427	0.0509	-	0.991
κ	$\text{cm}^2 \text{ s}^{-1}$	$\kappa = \frac{a \cdot \theta_v}{b + \theta_v} + c$	0.0012	0.0276	0.00014	0.957
C	$\text{MJ m}^{-3} \text{ }^\circ\text{C}^{-1}$	$C = a \cdot \theta_v + b$	3.8772	0.5671	-	0.990

The bulk volumetric heat capacity of manure at the saturation point was determined as approximately $3.95 \text{ MJ m}^{-3} \text{ }^{\circ}\text{C}^{-1}$, close to that of water ($4.18 \text{ MJ m}^{-3} \text{ }^{\circ}\text{C}^{-1}$). This is due to the high porosity of dairy manure (≈ 0.90). Similar to the relationship between λ and θ_v , there is a strong linear relationship between C and θ_v . The bulk volumetric heat capacity of manure was determined based on the measured λ and κ . The small change of κ during the entire manure drying process (approximately within $0.0003 \text{ cm}^2 \text{ s}^{-1}$) did not significantly modify the regression form between λ and θ_v . As a result, the regression form identifying the relation between C and θ_v was identical to the expression for λ and θ_v . These results were in agreement with previous thermal property determinations, indicating the linear regression between C and θ_v (Ahn et al., 2009; Nayyeri et al., 2009; Opoku et al., 2006; Yang et al., 2002).

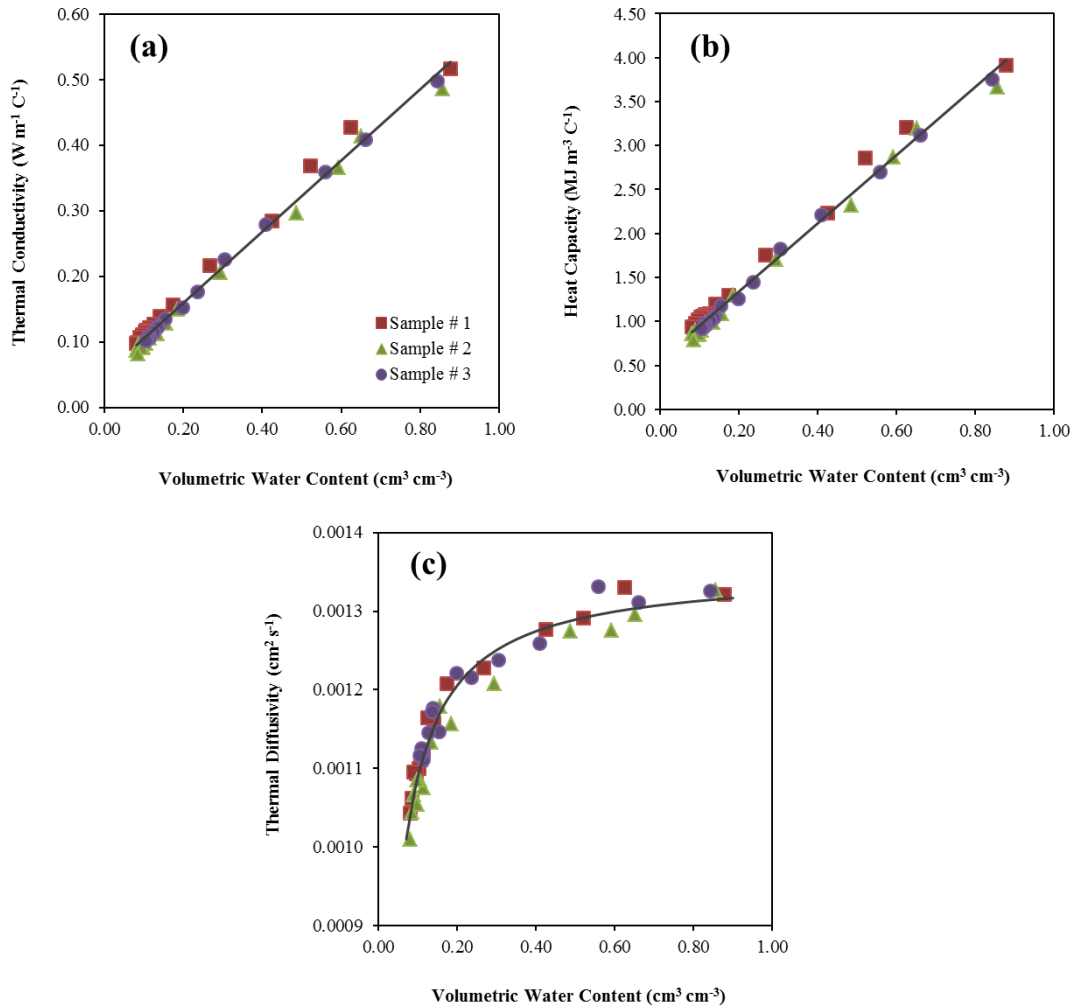


Figure 7. Changes in (a) thermal conductivity, (b) bulk volumetric heat capacity, and (c) thermal diffusivity with volumetric water content of dairy manure samples. The solid lines show the parametric expressions with the parameters indicated in Table 4.

While λ and θ_v are strongly linearly correlated, the relation between κ and θ_v was well approximated with the “plateau” curve expression as depicted in Figure 7(c). The parametric expressions including parameters are listed in Table 4. The change of κ during the manure drying process covers only a small range, approximately from 0.0013 (saturation) to 0.0010 $\text{cm}^2 \text{s}^{-1}$ (dry). Perhaps it is because of this narrow range that previous studies have failed to identify a specific relationship between κ and θ_v , where these include ascending, descending, and mixed trends (Bristow, 1998; Iwabuchi et al., 1999; Labance et al., 2006; Opoku et al., 2006).

Table 5. Summary of thermal properties of dairy and beef cattle manure reported in previous studies.

Medium type	Thermal properties			Observation ranges	Authors
	λ [W m ⁻¹ °C ⁻¹]	κ [cm ² s ⁻¹]	C [MJ m ⁻³ °C ⁻¹]		
Beef cattle manure	0.22 – 0.03	0.00150 – 0.00142	–	Wet weight basis between 100% and 0%	Houkom et al.(1974)
Beef cattle manure	0.63±0.059 – 0.064±0.003	1.515 – 0.849	–	TS† between 2.8% and 95% at 20 °C	Chen (1983)
Dairy manure	0.64 – 0.54	–	–	TS between 0% and 8%	Achkari-Begdouri and Goodrich (1992)
Mixture of fresh dairy feces and cedar sawdust	0.202 – 0.05	0.00155‡	–	θ_v between 0.44 and 0	Iwabuchi et al.(1999)
Dairy manure	0.55 – 0.08	0.001125 – 0.001250	–	Wet weight basis 80% and 20% at 40 °C	Nayyeri et al. (2009)
Beef manure compost	0.39 – 0.05	0.0010 – 0.0008	3.82 – 0.59	From saturation to dry conditions	Ahn et al. (2009)

† TS = Percentage of total solid (wet weight basis)

‡ Mean value, reported as independent of θ_v

The accuracy of the PHPPs in estimating the thermal properties was assessed through a comparison of θ_v derived from Eq. (6) with θ_v measured by the GS3 sensors. The specific heat capacity of solid organic matter was taken as 1,925 J kg⁻¹ °C⁻¹ (Hillel, 1998; Jury and Horton, 2004) and the dry bulk density was 0.15 g cm⁻³ (90% porosity). The regression relationship (Figure 8) suggests strong agreement between θ_v estimates derived from the GS3 sensors and those obtained with the PHPP method ($R^2 = 0.944$ and $RMSE = 0.0524 \text{ cm}^3 \text{ cm}^{-3}$). As illustrated in Figure 8, the regression equation indicates a slight bias towards greater overestimation of θ_v by the PHPP method at lower water content. However, hypothesis tests show that the difference of θ_v obtained by both methods was found to be insignificant ($p\text{-value} = 0.362$). This well-correlated relationship verifies the accuracy of the thermal properties determined by the PHPPs and the dry bulk density of dairy manure presented in this study.

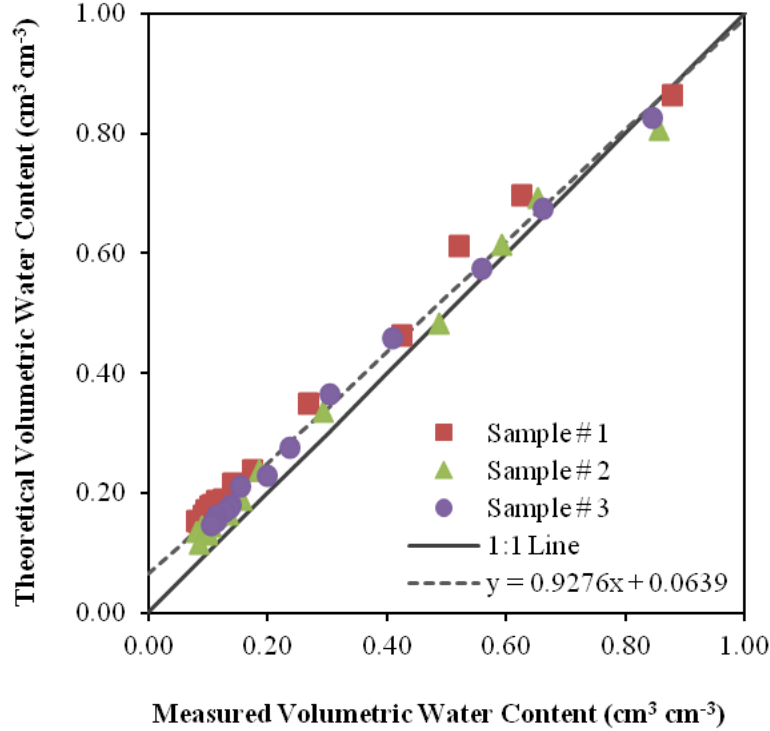


Figure 8. Theoretical volumetric water contents estimated from the heat capacity (C_m) using the PHPPs compared with the volumetric water contents measured with the GS3 sensors.

5. Summary and Conclusions

This study focused on the fundamental physical, hydraulic, and thermal properties of dairy manure that primarily affect the transport of liquid water and gas within the manure. Numerical modeling of transient water flow in cattle manure requires an accurate estimation of a number of physical and hydraulic parameters, including the water retention characteristic, saturated hydraulic conductivity, and the unsaturated hydraulic conductivity function. Measurement techniques commonly applied in soil science were applied to determine physical properties of as-excreted dairy manure, including the dielectric permittivity (K_a) - volumetric water content (θ_v). The uncertainties of the measurements were anticipated from the shrinkage phenomenon during the drying process. The liquid water retention characteristic for cattle manure, determined based on gravimetric measurements and the chilled-mirror dewpoint technique, was found to be similar to that of organic peat soils. Inverse analysis of the unsaturated hydraulic conductivity, using the developed water retention characteristic and laboratory evaporation experiment, yielded reasonable results, demonstrating strong support for the hypothesis that Richards equation can describe hydrodynamic processes taking place in dairy manure relevant to natural drying processes. The effects of surface crust formation and shrinkage, which are likely to occur variably upon drying, potentially modify the water retention and hydraulic conductivity functions due to high moisture content and high porosity of as-excreted manure. Further work is needed to characterize the manure's surface crust formation to more completely understand key processes

(e.g., gas emissions, nutrient leaching) impacting the environment and leading to a more sustainable system.

Thermal conductivity (λ), thermal diffusivity (κ) and bulk volumetric heat capacity (C), were determined during the course of manure drying, using PHPPs. Thermal properties of λ and C exhibited strong linear correlation with decreasing θ_v . Although κ also decreased with decreasing θ_v , it showed a more complex regression form. The accuracy and agreement of the thermal properties determined was assessed. The results suggested a reliable prediction of θ_v using the PHPPs, indicating well-estimated physical and thermal properties of dairy manure. The resulting thermal properties of dairy manure are likely to be used for development of heat transport models to identify the optimal conditions for manure composting processes as well as for prediction of manure water content and the movement of solutes and water from manure sources in addition to microbial activity and gas generation. Overall, the results presented here provide a solid foundation upon which future research can build in better modeling and understanding dairy cow manure processes that impact the environment.

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